Block Seminar: Machine Learning for Natural Language Processing

'A Survey on Transfer Learning'

by

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Abstract

As technology progresses and we make advents in data processing and analysis, there is a need to keep up with the ever-increasing amount of data and come up with ways to deal with learning scenarios that come into picture with the availability and categorisation of such data. One such field that has gained popularity over the years is machine learning. Supervised learning methods still constitute one of the best approaches to deal with learning scenarios. Another emerging field of research that has started to gain popularity over the past few years is that of transfer learning. While it is not a new discovery, the developments in deep learning and newly forming data needs and requirements have created a niche for transfer learning and induced a huge scope for research. This report is based on a survey paper that discusses the basic definitions and terminologies with respect to transfer learning, along with some lateral research done in different settings and approaches pertaining to it. For a novice, it covers basic ideas and notations about transfer learning. It also shows an overview of the progress made in this field over time through various research methodologies. It primarily states the role transfer learning techniques play in the area of Natural Language Processing. Further, it explores a bit into the challenges in transfer learning and scope for future research here.

Index Terms: Machine Learning, Transfer Learning, Inductive, Transductive, Unsupervised, Negative Transfer, Homogeneous learning, Heterogeneous learning, Instances, Features, Relations, Parameters

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Chapter 1

Introduction

At the basic level, machine learning is a technique of deriving and understanding complex patterns from data and using this knowledge to make predictions on unseen data. In computing terms, a computer is trained to recognise patterns from examples rather than implementing a specific set of rules in a program. For instance, a machine learning algorithm can be trained to differentiate between texts written in different languages based on the underlying patterns like grammar, syntax, script etc. Machine learning is applied to the tasks of classification, regression, or clustering in most common applications. The approaches for training a model are broadly divided into three types - supervised, unsupervised, and semi-supervised. The standard steps followed in any generalised model training process are data collection, pre-processing (feature extraction), choice of model, model training, evaluation, fine-tuning, and testing. For every new task at hand, these steps are followed from end-to-end in traditional machine learning scenarios. For example, for classifying images of flowers into different species and images of dogs into different breeds, while the underlying motive is the same, a traditional ML pipeline has to start from scratch and go through all the steps for achieving these objectives. Because the training is based on patterns and relations between the data, it becomes very important to correctly accumulate the data that a machine learning algorithm is trained on vis-à-vis the following:

1. Amount of data - The more the available data for training, the more precise the model is.
2. Relevance of data - The higher the relevance of the data to the problem statement, the more accurate the model becomes on training.
3. Variance and uniformity in features - The features extracted must all be related to each other and the problem at hand.
4. Distribution of training and test data - A model must be trained on the same sort of data on which it is supposed to make predictions.
5. Choice of model - Depending on whether the problem falls under supervised, unsupervised, or semi-supervised learning, different model choices like SVMs, Neural Networks, Naive Bayes etc. are available for learning.
6. Choice of performance metric/Objective function - Different loss functions are paired with different machine learning algorithms to enhance the performance of the model e.g. squared loss with linear regression, logistic loss with logistic regression.

Hence, most machine learning algorithms work under one common assumption that the training and test data have to be drawn from the same feature space and distribution. When there is a difference in the distributions, the models have to be retrained accordingly. Naturally, this is time-consuming, expensive, and sometimes not even feasible. Consider the flowers and dogs classification example. If the knowledge gained during training the model for classifying flowers can somehow be leveraged in some way to reduce the efforts for training over the same steps for classifying dogs, it makes the task at hand much easier. This is where transfer learning comes into picture.

Chapter 2

Transfer Learning: A Background

The motivation for research in transfer learning is believed to have come from the Neural Information Processing Systems (NIPS) 1995 Workshop called *"Learning to Learn: Knowledge Consolidation and Transfer in Inductive Systems"*. This workshop focused on the need for lifelong machine learning methods which would not only be able to retain, but also reuse the previously obtained knowledge. Almost two decades later in 2016, in another NIPS conference, Andrew Ng, a renowned name in the field of Machine Learning, declared in his tutorial that "After supervised learning, transfer learning will be the next driver of ML commercial success."

What drives transfer learning is a contradiction to the underlying assumption about machine learning - that domains, tasks, data, and feature distributions used in training and testing have to be the same. Transfer learning allows these to be different. This will help us delve into the meaning of transfer learning. Here are a few handpicked definitions of this term:

Definition 1: *"Transfer Learning (TL) is a research problem in machine learning (ML) that focuses on storing knowledge gained while solving one problem and applying it to a different but related problem."* - Wikipedia

Definition 2: *"Transfer Learning is a situation where what has been learned in one setting is exploited to improve generalisation in another setting."* - Deep Learning (Goodfellow et al.)

Definition 3: *"Transfer or Inductive Learning is a supervised learning technique that reuses parts of a previously trained model on a new network tasked for a different but similar problem."* - Deep AI

We will look at a more technical definition later when we see some basic terms and notations. But what transfer learning essentially means is to transfer the knowledge about doing something in order to do something else. Basically, there is no need to learn from scratch in order to solve the task.

Humans often implement transfer learning in their daily tasks without even realising it. Children who are once shown an apple will be able to recognise apples in any other size, shape, or location. Knowing how to ride a bicycle makes it easier to learn to ride a motorbike. Another example is where one generally has to learn Object Oriented Programming concepts to work on a language like Java. When one picks a new language like Python for learning, the knowledge of these concepts in Java makes it easier to learn the implementation of classes and objects in Python without having to learn the concepts from scratch. Transfer learning is also the ability of a system to recognize and apply knowledge and skills learned in previous tasks to novel tasks (in new domains). Over a span of years, a lot of research has gone into this field, and today, this term goes by different names like 'Incremental/Cumulative Learning', 'Learning to learn', 'Lifelong learning', 'Knowledge transfer', 'Inductive transfer', 'Knowledge consolidation', 'Context sensitive learning', 'Knowledge based inductive bias', 'Meta learning' etc.

However, transfer learning cannot be applied everywhere. Indeed, there are scenarios where applying transfer learning can adversely affect the performance. This is known as negative transfer, and will be discussed in detail in section 9. Thus, applying transfer learning techniques to a problem raises four questions:

1. To transfer or not to transfer - Transfer learning can adversely affect the efficiency and performance of the machine learning model, so researching whether it is applicable to the task at hand is an important step. An image classification model cannot be used for text prediction, and transfer learning will not help here.
2. What to transfer - Not all knowledge is useful. Some knowledge is task specific, while some knowledge is common to different tasks. While mathematics is a prerequisite for understanding machine learning, trigonometric functions will most likely not help in learning ML concepts the way calculus, probability and statistics will. So it becomes important to determine what features and "knowledge" are transferable to learn the task.
3. How to transfer - This brings into picture the various algorithms and methods studied based on the kind of problem and availability of data. The paper, "A Survey on Transfer Learning"[1] on which this report is based mentions these in great detail.
4. How much to transfer - Knowing when to stop transferring is important in the interests of data and time constraints, and also to avoid overfitting the model.

This report is restricted to covering transfer learning techniques applicable to the tasks of classification, regression, and clustering only. It serves an introductory purpose to get the reader acquainted with a broad level view of different transfer learning techniques, methodologies, research, and challenges. Sections 3 and 4 discuss the basics behind why transfer learning is useful and how it differs from traditional Machine Learning. The terminologies that will be used in the rest of the paper while discussing this topic are covered in Section 5. Sections 6 and 7 discuss the different scenarios encountered in transfer learning. A literature review of the paper on which this report is based is briefly discussed in Section 8. Lastly, sections 9, 10, 11, and 12 discuss the recent advances made in this field with the advent of techniques like deep learning, how the field of NLP has been impacted by it, its applications, the challenges and future scope involved, followed by a brief summary of the discussion.

Chapter 3

Motivation

Data, speed, and performance are the cornerstones of a well-functioning machine learning algorithm. The need of machine learning is to be able to solve complex problems, but the caveat is that models that solve complex problems need a lot of data. Such datasets do not exist in most cases. The large data repositories available today on various platforms have been amassed through months of meticulous collection efforts. Even if data is available, it is hardly ever of any use in its coarse, unstructured, and unlabelled format. Most models need structured, pre-processed, and labelled data for efficient training. There is also the question of how long the data remains relevant, because outdated data is of no use to the model. For example, for a company that wishes to assess the employee satisfaction level of its current employees, the database of employees from ten years ago is of no use. There are cases where training data is expensive or difficult to collect, rare, or even private and inaccessible. Transfer learning helps to overcome all these data related issues. We shall see how in the further sections.

Another advantage of transfer learning is that because it harnesses pre-trained model information, there is a marked improvement in the speed of training.

It has also been observed that in situations where transfer learning is applicable to the problem statement, there is an increase in the model performance as compared to model training without transfer learning, as seen in Fig. 3.1.

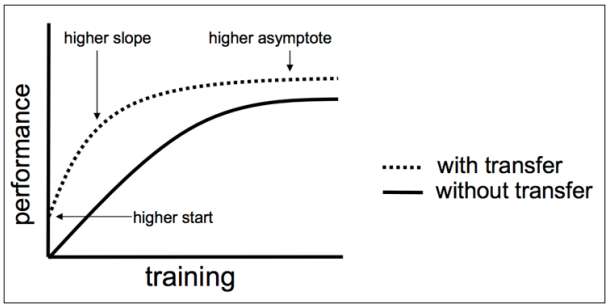


Figure 3.1: source: https://machinelearningmastery.com/transfer-learning-for-deep-learning/

Today, strides in Big Data analytics have made it possible to have easily available big data repositories that can be used as baselines for training other models pertaining to the same domain. A few such examples are ImageNet for Computer Vision, Word2Vec for Natural Language Processing, knowledge bases like Wikidata, YAGO etc.

Thus, the ultimate motivation behind transfer learning is to leverage knowledge from pre-trained models, tackle the problem of data sparsity, and increase the overall performance of the model.

Chapter 4

Traditional Machine Learning vs Transfer Learning

Fig. 4.1 shows the differences between a traditional Machine Learning system and a Transfer Learning system. As one can see, traditional machine learning involves learning each task from scratch, while in transfer learning, one utilises the knowledge from previous tasks to a target task. In the target class, there is generally a small amount of training data available.

Consider the previously introduced example of classifying images of flowers and dogs into their respective categories. In traditional Machine Learning, we would view these as two separate unrelated problems. Two separate classifiers would be trained from scratch, one with images of flowers, and the other with images of dogs. These would be two completely separate, isolated systems. Now, we can easily see that both are primarily image classification tasks which have to be able to classify the images by learning the underlying patterns and objects in the images. Using transfer learning, it can be possible to train a classifier of flowers, and modify this classifier to be able to classify dog images. When we say 'underlying patterns and objects' we're talking in essence regarding the knowledge being transferred from one model to another. In Computer Vision, this shared knowledge is the image features, corners, edges, shapes, intensity etc. In tasks related to Natural Language Processing, the transferred knowledge could be the underlying grammar, syntax, word structure, common roots of different languages etc. In traditional ML, if one wants to build a POS Tagger for the German language, one would need the underlying data and feature set to be from the same domain i.e. the German language. Using transfer learning, English language taggers can be leveraged to train such a POS Tagger.

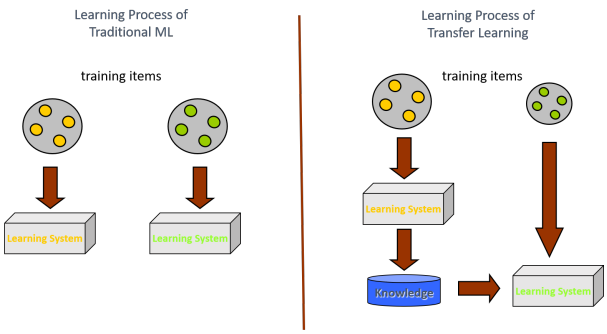


Figure 4.1: Source: 'A Survey on Transfer Learning', Sinno Jialin Pan and Qiang Yang, IEEE

Having understood the meaning of transfer learning and motivation behind it, we shall now deep dive into some technical terms and notations.

Chapter 5

Notations and Terminologies

In this section, a few notations and terminologies are introduced which shall be used in the rest of the report. We will define a few standard terms as follows:

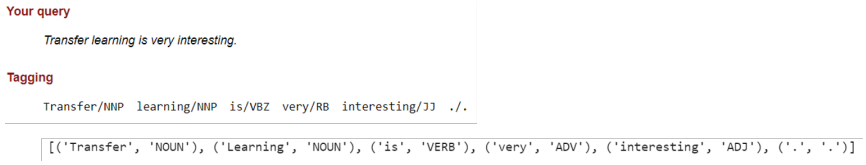
- Feature Space - Feature space refers to the n-dimensions of the variables of the data other than the target label, where all the variables are known as features. An n-dimensional variable X is composed of the features $\{x_1, x_2, x_3, ..., x_n\}$.
  Edges, corners, ridges, pixel intensity would be features in an image classification problem.
- Label/Class - A label is the category a particular target variable is assigned to.
  In the task of classifying images of dogs, the class labels would be 'Labrador', 'Retriever', 'Pug', 'Bulldog' and so on.
- Marginal Distribution - Marginal distribution of a variable X is the probability of that variable occurring in the feature space.
- Conditional Distribution - The conditional distribution of a class Y given a variable X is the probability distribution of Y if the value of X is known.
- Domain - A domain is a tuple of the form $(X, P(X))$, where X is the feature space and *P(X)* is the marginal distribution.
- Task - A task is a tuple of the form $(Y, P(Y|X))$ which is the label and the predictive function and can be denoted by $T = \{Y, f(.)\}$. It is not known, but is learned from the training data, which consists of pairs $\{x_i, y_i\}$.

To understand these terms, we shall take the example of document classification. The document is represented in a feature space $X = \{x_1, x_2...x_n\}$, which can just be a Bag-of-Words model, with x_i being the ith term vector. The label space Y contains the document class labels such as {*'History', 'Maths', 'Science', 'General Knowledge'...*}. The marginal distribution is the probability of the term x_i occurring in the document, while the conditional distribution is the probability of a document D_i having the label y_i. The classification task finds out the probability of D_i having label y_i using $P(y_i|x_i)$ for all x_i word terms in document D_i.

In transfer learning, we will only consider simplistic scenarios of knowledge transfer between one trained model, referred to as the source model, and the model to be trained, referred to as the target model. Which brings us to more specific terminologies:

- Source Domain D_s - All tuples $\{(x_{s1}, y_{s1}), (x_{s2}, y_{s2}), ...(x_{sn}, y_{sn})\}$ where $P(X_s)$ is the marginal distribution of X_s.
- Source Task T_s - $P(Y_s|X_s)$ where $Y_s = \{y_{s1}, y_{s2}, ..., y_{sn}\}$
- Target Domain D_t - All tuples $\{(x_{t1}, y_{t1}), (x_{t2}, y_{t2}), ...(x_{tn}, y_{tn})\}$ where $P(X_t)$ is the marginal distribution of X_t.
- Source Task T_t - $P(Y_t|X_t)$ where Y_t is the target label space.

(a) Representation of number 5 in different domains



(b) Labels of two Different POS Taggers

Figure 5.1: An illustration of different features spaces and target outputs

In transfer learning, we want to determine $P(Y_t|X_t)$ given D_S, T_s and D_t. Using all these terminologies and notations helps us to come up with a more unified, technical definition of transfer learning which is as follows:

Definition: 'Given a source domain D_s and learning task T_s, a target domain D_t and learning task T_t, transfer learning aims to help improve the learning of the target predictive function $f_t(.)$ in D_t using the knowledge in D_s and T_s, where $D_s \neq D_t$, or $T_s \neq T_t$'

In the above definition, when we say that the source and target models are not the same, there are four ways they can be different. We list below a brief explanation of each kind of difference along with the corresponding interpretations for a Natural Language Processing task.

1. $X_S \neq X_t$ - The source and target have different feature spaces.
   NLP interpretation: Documents are in different languages (Cross-lingual adaptation). It could also mean that a feature can be represented in different formats, as shown in Fig. 5.1a.
2. $P(X_S) \neq P(X_t)$ - The source and target have different marginal distributions.
   NLP interpretation: Documents discuss different topics, and so the term frequency of each word differs across the documents (Domain adaptation). This is called frequency feature bias. For example, consider health related info and crime reports. Both will have a different distribution of features like number of adjectives in the text, or number of occurrences of words like "hospital" and "prison".
3. $Y_S \neq Y_t$ - The source and target have different labels spaces.
   NLP interpretation: The documents are assigned different labels. For example, the NLTK and Stanford POS Taggers both carry out the same tasks, but use different tagsets for classification (See Fig. 5.1b).
4. $P(Y_S|X_S) \neq P(Y_t|X_t)$ - Source and target documents are unbalanced w.r.t their classes.
   NLP interpretation: The same term can have different contexts in the source and target domains. This is knows as context specific bias. For example, the word 'address' will have a different meaning in the context of a post office and a politician's rally.

In standard transfer learning problems, one or more of these differences hold true. Of course, if all these things are the same, it means this is just one ML problem, and there is no need to do transfer learning at all!

Chapter 6

Settings of Transfer Learning

According to the definition of transfer learning, we can classify it under 3 settings based on the availability of data, type of problem, and the difference in the source and target space. Fig.10.2 gives a clearer understanding of these settings.

1. Inductive Transfer Learning - In this setting, the target task is different from the source task (the domains could be same or different). Some labelled data is available in the target domain. This labelled data is used to *induce* a predictive model function $f_t(.)$ in the target domain. Based on the source data, we can sub-categorise it into 2 types:

   - The source has a lot of labelled data: This is similar to a traditional ML multi-task learning scenario. However, in multi-task learning, multiple models learn simultaneously, but in transfer learning, the goal is to achieve a high performance in the target task.
   - The source has no labelled data: Similar to self-taught learning, the source information cannot be used directly for learning, because the source and target label spaces do not match (in this case, source label space is non-existent).

2. Transductive Transfer Learning - The term transductive transfer learning was first proposed by Arnold et al.[2]. Here, the source and target tasks are the same, but the domains are different. Domains being different indicates that either $X_s \neq X_t$ or $P(X_s) \neq P(X_t)$. Different marginal distributions is a similar case to domain adaptation. A sample selection bias or covariate-shift assume that the source data is sampled from the target data and so they both have the same distribution. To get a clearer understanding of the difference between inductive and transductive learning, consider you have $\sim$ 2000 labelled instances and $\sim$ 50000 unlabelled ones. Inductive learning will utilise the 2000 instances to build a classifier, while transductive learning will state that even if the 50000 instances are unlabelled, they can be used to get more information about the problem space.

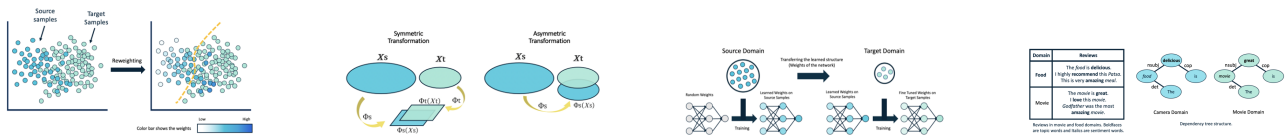
3. Unsupervised Transfer Learning - In unsupervised learning, neither the source not the target spaces have labelled data, but the target task, though different from the source task, is related to it. This setting is used for solving tasks like clustering, dimensionality reduction etc. in the target domain.

Chapter 7

Approaches to Transfer Learning

Based on the type of task and availability of data, we categorise transfer learning into the above mentioned settings. The next question to answer is "What to transfer?" and this helps us to classify transfer learning techniques into the following types (refer to Figure 7.1 for illustrations):

1. Instance Based - This approach assumes that certain parts of data in the source domain can be used for learning in the target domain by *re-weighting*. Since available data is sparse in the target domain, we try to use some of the source data along with the target data to train the target model. Source data that is more similar to the target data is assigned a greater weight (Fig 7.1a).
2. Feature Based - The intuition behind this approach is to find commonalities between the source and target domains and learn a "good" common feature representation to reduce the difference between them as well as lower the classification/regression error. This feature transformation can be (ref. Fig 7.1b)
   a) Symmetric i.e. both, source and target feature spaces are transformed
   b) Asymmetric i.e. only the source feature space is transformed
3. Parameter based - The assumption behind this approach is that the source and target share some parameters or common prior distributions, using which the knowledge transferred can be encoded in the shared parameters. The model trained on the source task has a well-defined structure, and if the two tasks are similar, this structure can be transferred to the target model. A common example of this is retraining the last few layers of a pre-trained neural network to use for another model (Fig 7.1c).
4. Relation based - This approach is useful when the data is not i.i.d. When the relationship between the data is similar for the source and target domains, such knowledge can be transferred by building a mapping of the relational knowledge.



(a) Instance based (b) Feature based (c) Parameter based (d) Relation based

Figure 7.1: source: https://medium.com/georgian-impact-blog/transfer-learning-part-1

	Inductive Transfer Learning	Transductive Transfer Learning	Unsupervised Transfer Learning
Instance Transfer	✓	✓	
Feature Representation Transfer	✓	✓	✓
Model Transfer	✓		
Relational Knowledge Transfer	✓		

Figure 7.2: Approaches in Transfer Learning

Table 7.2 shows which approach has been studied under which setting according to the paper by Sinno Jialin Pan and Qiang Yang[1].

Chapter 8

Literature review

This report is largely based on the paper "A Survey on Transfer Learning" by Sinno Jialin Pan and Qiang Yang Fellow in IEEE, '09. It focuses on categorising and reviewing the progress in transfer learning for classification, regression, and clustering problems. A comparative study has been made between research in transfer learning and other related machine learning techniques such as multi-task learning, domain adaptation, sample selection bias etc. The paper acts as a resource for comprehending the basics of transfer learning and also describes the studies conducted in this field. The definitions introduced in this survey have been used and cited by many other articles and papers written later on Transfer Learning.

We have already seen the terminologies, notations, settings, and approaches pertaining to transfer learning so far. This paper describes some methods and techniques studied in different approaches and settings of Transfer Learning, as highlighted by the checkmarks in Fig.7.2. We will briefly summarise some of the undertaken research approaches, as listed in the survey.

1. Inductive Transfer Learning

   (a) Instance based transfer - Approaches like boosting, conditional probability differences, heuristics, and SVMs have been studied for instance based learning. The algorithm TrAdaBoost[3] has been used as a baseline for further research with the presumption of using boosting to reduce the effect of the "bad" source data and encourage the "good" source data to contribute to the domain.

   (b) Feature based transfer - Supervised and unsupervised feature construction methods that use optimisation techniques, kernel based SVM approaches, manifold learning etc. have been described for feature based learning. Supervised feature construction tries to learn low level features, while unsupervised methods use sparse data and try to find the higher level features. The aim is to reduce the domain divergence and classification/regression error of the model.

   (c) Parameter/Model based transfer - Parameter based learning has seen a lot of research via techniques like Gauss based priors, Hierarchical Bayesian frameworks, and ensemble learning methods. Regularised SVMs[4] have also been used here by considering that the learning parameter w for an SVM can be split into source specific v_s, target specific v_t, and common w_0 terms. These can be learned by using optimisation like regular SVMs.

   (d) Relation based transfer - In relational domains where the data is non i.i.d., statistical relational learning techniques have been proposed to solve this problem. TAMAR, an algorithm proposed by Mihalkova et. al[5] has exclusively researched the usage of Markov Logic Networks to map the relational data between the source and target domains. MLNs combine the graph-oriented approach of Markov Random Fields, and the expressive power of first order/predicate logic to represent the relationships. Similar techniques that use Markov models have been studied thereafter.

2. Transductive Transfer Learning

   (a) Instance based transfer - In the Transductive Learning setting, sample selection bias motivated by importance sampling has been mentioned to solve the problem of instance based transfer. Here, the properties of the target distribution are estimated using the available source data

distribution by finding out the marginal distribution ratio of the source and target by using methods like classification, kernel-mean matching, Naive Bayes, and KL divergence minimisation.

(b) Feature based transfer - Structural Correspondence Learning[6] is a useful way to determine pivot features that help to align the source and target feature spaces by using SVD or heuristic methods like mutual information. SCL identifies pivot features, removes them from the data and treats each pivot feature as a new label vector. Using these pivot features, a mapping that reduces the distance between the source and domain spaces is learned.
Features are also mapped using kernel mapping, co-clustering, bridged refinement, spectral classification, probabilistic latent semantic analysis, Maximum Mean Discrepancy, and Transfer Component Analysis by different research methods.

3. Unsupervised Transfer Learning

(a) Feature based transfer -The Unsupervised Learning settings works on the approach of transferring features by using Self-taught Clustering[7] of a small amount of unlabelled data in the target domain with the help of a lot of unlabelled data in the source domain. This is done by finding a common feature space across the domains and using it to cluster on the target domain. Another approach is to use a method called Transferred Discriminative Analysis[8], which applies clustering to generate pseudo-class labels for the target data and then uses dimensionality reduction on the source and target data.

The results of the methods mentioned in this paper have been illustrated in Fig. 8.1 as reported in the survey. It can be seen that transfer learning approaches certainly provide an improvement over the traditional machine learning baselines.

Data Set (reference)	Source v.s. Target	Baselines		TL Methods	
20 Newsgroups$_1$ ([6])		**SVM**		**TrAdaBoost**	
ACC (unit: %)	rec v.s. talk	87.3%		92.0%	
	rec v.s. sci	83.6%		90.3%	
	sci v.s. talk	82.3%		87.5%	
20 Newsgroups$_2$ ([84])		**SVM**		**TrAdaBoost**	**AcTraK**
ACC (unit: %)	rec v.s. talk	60.2%		72.3%	75.4%
	rec v.s. sci	59.1%		67.4%	70.6%
	comp v.s. talk	53.6%		74.4%	80.9%
	comp v.s. sci	52.7%		57.3%	78.0%
	comp v.s. rec	49.1%		77.2%	82.1%
	sci v.s. talk	57.6%		71.3%	75.1%
20 Newsgroups$_3$ ([49])		**SVM**	**LR**	**pLWE**	**LWE**
ACC (unit: %)	comp v.s. sci	71.18%	73.49%	78.72%	97.44%
	rec v.s. talk	68.24%	72.17%	72.17%	99.23%
	rec v.s. sci	78.16%	78.85%	88.45%	98.23%
	sci v.s. talk	75.77%	79.04%	83.30%	96.92%
	comp v.s. rec	81.56%	83.34%	91.93%	98.16%
	comp v.s. talk	93.89%	91.76%	96.64%	98.90%
Sentiment Classification ([8])		**SGD**		**SCL**	**SCL-MI**
ACC (unit: %)	DVD v.s. book	72.8%		76.8%	79.7%
	electronics v.s. book	70.7%		75.4%	75.4%
	kitchen v.s. book	70.9%		66.1%	68.6%
	book v.s. DVD	77.2%		74.0%	75.8%
	electronics v.s. DVD	70.6%		74.3%	76.2%
	kitchen v.s. DVD	72.7%		75.4%	76.9%
	book v.s. electronics	70.8%		77.5%	75.9%
	DVD v.s. electronics	73.0%		74.1%	74.1%
	kitchen v.s. electronics	82.7%		83.7%	86.8%
	book v.s. kitchen	74.5%		78.7%	78.9%
	DVD v.s. kitchen	74.0%		79.4%	81.4%
	electronics v.s. kitchen	84.0%		84.4%	85.9%
WiFi Localization ([67])		**RLSR**	**PCA**	**KMM**	**TCA**
AED (unit: meter)	Time **A** v.s. Time **B**	6.52	3.16	5.51	2.37

Figure 8.1: Results against baselines

This paper also lists down challenges encountered in transfer learning, which have been described in detail in the subsequent section. However, reading this paper only provides a foray into the field of transfer learning. One cannot directly classify an ML task into one specific category as mentioned in the survey. This means that on being given a transfer learning problem, it will still be difficult to determine if the task falls perfectly into the niche of one particular setting and approach. Also, this paper is relatively old and does not cover the newer trends being researched, especially ones like Neural Networks and Deep Learning methods. A brief overview of these will be given in section 10.

Chapter 9

Challenges

Transfer Learning is not always best suited to any machine learning task. Sometimes, using transfer learning can actually degrade the performance of the system. This phenomenon, where something learned in one situation actually hinders the learning in another situation is known as **negative transfer**. Here, the source domain data and task contribute to the reduced performance and learning in the target domain. Negative transfer occurs when
a) source and target tasks are not related
b) the chosen transfer method does not correctly capture the relationship between the source and target tasks

An example of this is as follows. The English language generally uses the adjectives before the nouns, while in Spanish adjectives are after the nouns. Transfer learning here could negatively impact the learning of Spanish grammar from English based source rules. Some claims state that the root cause of negative transfer is mainly due to conditional distribution differences between source domains and a difference in class distribution (class imbalance) between the source and target spaces. Studies have shown that brute-force transfer may actually hurt the performance of the target task [9].

Hence, there is a need to determine how "transferable" a problem is from the source to the target. This transferability measure is defined as follows - transferability from the first learner to the second learner is the performance of the second learner with learning from the first learner minus the performance of the second learner without learning from the first learner. Spectral graph theory is used to derive this transfer function from one learner to another. Note that this function is not commutative i.e. the graph would not be bidirectional.

The amount of knowledge transferred can also be limited by transfer bounds for quantifying the transfer. Kolmogorov complexity[10] has been researched as one such bound. This is theoretically defined as the length of the shortest program that produces some kind of output. The analysis for this involves using the conditional Kolmogorov complexity to transfer the "right" amount of information in a Bayesian framework based sequential transfer learning task by finding out the relatedness. Other graph embedding methods[11] have also been utilised in this context. The studies in this area try to group tasks with the assumption that similar tasks would fall under the same group and transferring useful knowledge within them would have a positive impact.

Negative transfer, transferability metric, and transfer bounds are three important considerations that determine the performance of a transfer learning task. However, robust negative transfer measurements are difficult to define. Since the target domain typically has limited labeled data, it is inherently difficult to find a true measure of the relatedness between the source and target domains. Further, by selectively transferring information that seems related to the limited labeled target domain, a risk of overfitting in the target learner also becomes a concern.

How to overcome these challenges is still an ongoing field of research with a lot of further scope.

Chapter 10

Recent Research and Trends

The paper by Sinno Jialin Pan and Qiang Yang Fellow works as a foundation for understanding the principles of transfer learning. The main purpose behind it is to give a broader level perspective of different approaches and transfer learning techniques. It gives a small insight into the kind of research methodologies that have been studied and the advances made accordingly. However, since the publication of this paper in 2009, other research and developments have led to a slight restructuring and updated definitions and terminologies. In this section, we will focus on the newer interpretations of different terms and settings.

Thanks to the advances in deep learning, another taxonomy has become a part of the scope of transfer learning. This taxonomy categorises transfer learning problems into two classes based on the similarity of domains, regardless of the availability of labelled and unlabelled data. Recall that when we say two domains are different i.e. $D_S \neq D_t$, it means that $X_S \neq X_t$ and/or $P(X_S) \neq P(X_t)$.

1. Homogeneous Transfer Learning - In this case, $X_s = X_t$ and $Y_s = Y_t$. Homogeneous Transfer Learning solutions focus on bridging the data distribution gap between the source and target domains. This means

   (a) Trying to correct the marginal distribution difference $P(X_s) \neq P(X_t)$
   (b) Trying to correct the conditional distribution difference $P(Y_s|X_s) \neq P(Y_t|X_t)$
   (c) Trying to correct both, the marginal and conditional distribution differences

2. Heterogeneous Transfer Learning - In this case, $X_s \neq X_t$ and/or $Y_s \neq Y_t$. This means that the feature spaces are generally non-overlapping. Heterogeneous transfer learning solutions try to bridge the gap between the feature spaces to convert the problem into a homogeneous transfer learning problem where the other distribution differences are corrected. It is a relatively new field of study that tries to align the input spaces of the source and target domains.

Weiss et al. in their more recent survey on transfer learning[12] formulate these two terms and then mention the research conducted by building on previous methods as well as developing new ideas for building transfer learning solutions. They further classify these two "new" settings of transfer learning into the same approaches as mentioned in the literature survey - i.e. that into instance based, feature based, model based, and relation based methods. Both these techniques play an important role in the Big Data domain. If there is an available dataset that is drawn from a domain that is related to, but does not an exactly match a target domain of interest, then these transfer learning settings can be used to build a predictive model for the target domain.

Machine Learning methods like ensemble learning, feature-weighting, kernel methods, PCA, manifold based regularisation, graph theory, unsupervised clustering methods, boosting (an improvement on the aforementioned TrAdaBoost), and bootstrapping are mentioned to tackle the homogeneous transfer learning problem.

Heterogeneous solutions build on spectral mapping, SVM based optimisation, kernels etc. The game changer in ML - Neural Networks - are also briefly mentioned in this survey in the form of of stacked denoising autoencoders for learning invariant features.

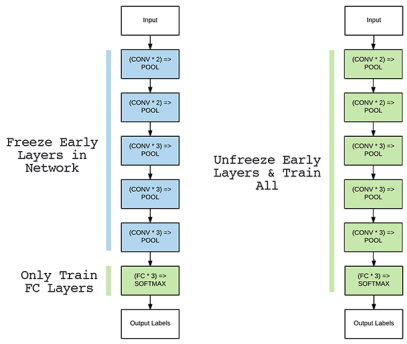


Figure 10.1: Transfer Learning with Neural Networks
(Source:https://www.pyimagesearch.com/2019/06/03/fine-tuning-with-keras-and-deep-learning)

There is a myth that deep learning requires a lot of labelled data to solve the problem. In reality, useful representations can be learned from unlabelled data and learned representations can be transferred from a related task.

Today, the most popular transfer learning applications leverage neural networks to transfer knowledge across the tasks. The neural net is trained on the source problem which learns different features at different layers. The lower layers learn very generic features like pixel data in images, while the upper layers learn features particular to the task. These last few "task specific" layers are replaced for the target model and only these layers are trained on the target task data. The model's pre-trained weighted layers are leveraged to extract features, but these layers are not changed in any way. This is called *freezing* the layers of the neural network. A variant of this approach utilises the pre-trained layers but *fine tunes* them by updating their weights to cater to the target task for better performance. Fine tuning is used when enough labelled target data is available to retrain the lower level layers too[13]. These approaches are illustrated in Figure 10.1.

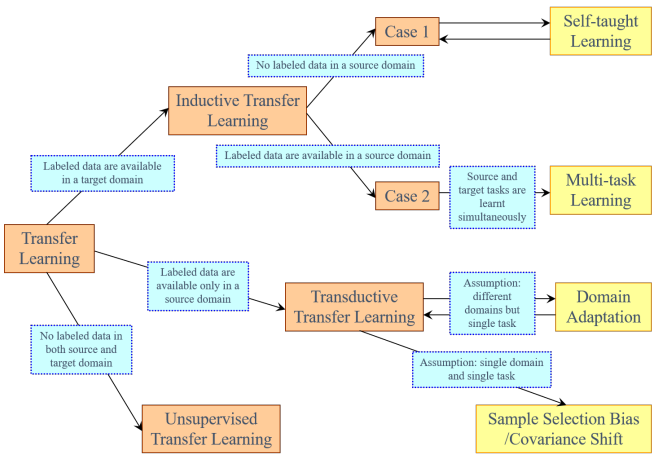


Figure 10.2: Settings in Transfer Learning

Chapter 11

Impact of Transfer Learning on Natural Language Processing

Until a while ago, applying transfer learning methods to the field of Natural Language Processing had not been researched a lot. While Computer Vision can harness plenty of the advantages offered by transfer learning, it was not easy to modify such settings to text and speech data the way it was for pixel data. This is because vector representations of images can be extremely dense. Those of texts, especially Bag-of Words models can end up being very sparse. They also do not capture the context and semantic relationships between the words in the documents, thus forming poor models.

The idea behind utilising transfer learning for NLP tasks is to get pre-obtained semantic word embeddings for data, and then use these for other tasks like translation, text classification etc. using transfer learning techniques. Classically, NLP tasks have been performed through rule-based and statistical methodologies. In recent times, state of the art models have been devised that can be used universally to leverage target tasks for different tasks in sentiment analysis, information retrieval and so on. This has also been termed as NLP's ImageNet moment, and a large contributor to this is the advances in neural networks.

A Neural Network trained on language corpora can be divided into two parts:-

- Language Specific: The lower part (layers) of the neural network is language specific. It captures the nuances of the natural language such as the low level features, syntax, symbol representations etc. This part is used to transfer knowledge from the language domain to the target task.
- Task Specific: The upper part (layers) of the network are task specific. The weights for these layers are learned from the target data for the corresponding prediction task.

One approach researched in this context is that instead of using a domain specific representation, it can make more sense to capture domain invariant features that can be applicable to all sorts of tasks. One way to ensure similarity between the representations of two such domains is domain confusion. This is done by adding an objective to an existing model that captures the domain classification loss instead of the label classification loss. A gradient reversal layer tries to maximise this domain classification error.

The more popular approach is to use some pre-trained models for Natural Language Processing such as Word2Vec, Glove, and FastText. Transfer Learning is at the heart of models like ELMo (Embeddings from Language Models)[14] and BERT (Bidirectional Encoder Representations from Transformers)[15] which aim at forming *universal embeddings* trained on a huge corpus (similar to ImageNet). However, in the sense of a generative context, a lot more research needs to be put in to apply lower level features from a trained neural network, as this seems to have better results in the field of Computer Vision[16].

One problem with such universal embeddings is that there is a large amount of data available in the English corpus, but other languages are not as well represented. These models also rely on massive hand-labelled datasets for training. Keeping these models up to date with the changing language contexts in the real world is also expensive.

Chapter 12

Summary and Conclusion

In this report, we get a basic understanding of what transfer learning is and how it differs from traditional machine learning. We study three settings and four different approaches transfer learning problems fall under. These settings are inductive learning, transductive learning, and unsupervised learning. The approaches on "what-to-transfer" are instance based transfer, feature based transfer, parameter or model based transfer, and relational knowledge based transfer of data. We also see how the recent terminology classifies these settings into two types - Homogeneous and Heterogeneous Learning. We get a comprehensive overview of the similarities and differences in these scenarios. Then we understand the challenges and research issues that are associated with transfer learning. We understand what negative transfer is and how important it is to avoid it. We come across the concept of "transferability measure" which determines the amount of useful knowledge to be transferred from the source domain to the target domain to induce effective learning without the risk of underfitting or overfitting.

After this, we delve a bit into the more recent research done in transfer learning.

Despite the challenges and shortcomings of transfer learning, its advantages largely outweigh the cons. Hence, it has been successfully applied to a variety of real-life scenarios.

1. Real World Simulations - In fields like robotics and automated driving, where training an agent in the real world is time-consuming, costly, and also risk-prone, agents are trained in simulated environments and this knowledge is transferred to real world data.
2. Games - With transfer learning, the tactics learned in a game can be reapplied to play another game. A combination of transfer learning and reinforcement learning proves much more effective on game playing tasks.
3. Image Processing - As explained in this report, the field of Computer Vision has made strides due to the advances in transfer learning techniques. They help reduce the time taken to train the model by pre-training the model using datasets like ImageNet, which contains millions of images from different categories. This has proved especially beneficial in fields like medicine.
4. Zero Shot Learning - The goal here is learning to predict novel values from values that are not present in the training dataset[17]. e.g. In a non-zero shot environment, two discrete languages have to be translated using a pivot language (e.g. Spanish to English to Japanese). This can result in distortions created by the dual way translations. Transfer learning zero-shot methods use available training data to learn the translational knowledge to translate a new language pair, and thus eliminate the need for a pivot language.
5. Sentiment Classification - The widest application of transfer learning for sentiment classification is in review systems for user feedback and product reviews. This involves training the models on any one domain and fine-tuning them to another domain to perform sentiment analysis on.

Besides these, Transfer Learning also finds applications in tasks like text classification, named entity recognition, WiFi Localisation, spam filtering and so on.

Thus, we see the impact transfer learning methods have had in Natural Language Processing and the subsequent advancements made due to this.

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